

A ternary finite network for integer multiplication

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Abstract

We extend the rational-frame interchange transfer to a fixed finite scalar alphabet and construct a ternary five-subset network with shared common-pair side computations. Every auxiliary role carries arbitrary data and is restored. The finite circuit supports bit saving $467/10^{11}$. Combined with the retained complex network, compact-control movement and precision proofs, it gives the conditional exponent $\kappa = 2^{-30}$. The upstream algorithmic claims remain assumptions.

The upstream manuscript is pinned at

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

This result uses the same fixed finite-alphabet, fixed-tape machine model. It makes no practical runtime or unrestricted optimality claim.

Related submissions and attribution. Zhihao Chen (`jacklightChen`) submitted the same ternary five-subset motif and finite-alphabet direction earlier in PR #7, with a different compressed producer and a stronger claimed witness. The present note records a separate implementation and a weaker conditional checkpoint; it makes no priority claim for that motif. The related complex compression submission by `eumemic` in PR #3 is related work. The release includes its own retained complex construction, not an adoption or verification of those pending PRs. See the repository's contribution-review index for the dependency review queue.

1 A fixed finite scalar alphabet in the interchange transfer

The rational-frame interchange interface does not intrinsically require XOR as its pointwise scalar operation. The following extension concerns the swap interface only; it does not change the complex phase interface. All conclusions remain conditional on the upstream stream procedures.

Proposition 1. *Let $\mathcal{A}$ be a fixed finite alphabet containing distinct symbols $0, 1$. Suppose a fixed finite circuit on W roles consists of fixed pointwise maps on symbols in $\mathcal{A}$ and has, on every input in $\mathcal{A}^W$, the complete scalar action of a role permutation ρ . Assign the same rational m -by- m matrix to all incidences of each gate. If*

$$M_{\text{out}(\rho(w))} - M_{\text{in}(w)} = I_m, \quad s = \sum_e \text{rank}_{\mathbb{Q}}(M_{\text{head}(e)} - M_{\text{tail}(e)}) < Wm,$$

then the upstream power-width interchange proposition holds with these m, W, s and every rational $\tau \in (0, 1)$ satisfying $s/W < m^\tau$. It acts on bit data as well as arbitrary $\mathcal{A}$ -valued data.

*Prepared with assistance from OpenAI Codex. This is not independent review or endorsement by OpenAI.

Proof. Use the fixed tape alphabet $\mathcal{A}$ on the data track. Its symbols can alternatively be represented by a fixed number of bits per cell; this changes every scan cost by a fixed factor. For this induction let V count data entries, so a role stream has exactly V/W entries. For an original bit array this is its original logical volume. No conversion factor enters the recursive branching coefficient.

The elementary stream procedures move whole symbols and apply fixed pointwise maps to aligned symbols. The machine model already permits fixed finite alphabets and fixed tracks. All counters, descriptors and address radices remain as in the upstream proof. The prime used for rational address shears is chosen from their finite rational tables and is unrelated to any field used to name the data symbols.

Induct simultaneously for all $\mathcal{A}$ -valued arrays. Base-case scans permute symbols. Recursive matrix-shear calls therefore remain within the induction hypothesis even when a surrounding gate has produced symbols other than 0, 1. A common address permutation commutes with every fixed pointwise map, without any linearity assumption on that map. Consequently the same frame invariant gives

$$f'_{\rho(w)} = \Phi_{M_{\text{out}(\rho(w))}} \Phi_{-M_{\text{in}(w)}} f_w = \Phi_{I_m} f_w.$$

Merging the role streams undoes ρ , including on all scratch-role payload. The two other triangular updates convert this shear into the desired chunk interchange. The identical recurrence is

$$T(V, e) \leq \sum_{j=1}^s T(V/W, e/m) + O(V),$$

with constants depending on the fixed alphabet and circuit. Restricting initial data to 0, 1 is now legitimate: every completed recursive interchange restores those data values, while its interior may use other symbols. No invalid binary encoding or initialized scratch symbol is assumed.

The arbitrary-width reduction is unchanged as well. Its reserved high address digits provide at least W^k rows, using only $O(\log e)$ digits for the fixed m, W . Row padding increases volume by less than two, and radix padding by a fixed factor. Use the symbol 0 in padded entries; completed recursive calls are data permutations and therefore preserve the padded zero rows at their boundaries. Removing those rows and invalid address ranges consequently returns the original symbol array in the required order. No bound on the numerical size of W relative to m is added by this reduction. $\square$

In particular one may take $\mathcal{A} = \mathbb{F}_p$ for any fixed prime p . A signed exchange $(X, Y) \mapsto (-Y, X)$ is corrected to an unsigned exchange by negating the physical X bank at its final identity frame. This is a fixed pointwise map with zero added edge rank. The rational frames and all rank charges must still be checked; changing the scalar alphabet by itself supplies no improvement.

2 A ternary five-subset network

This section changes the finite scalar alphabet for the bit interchange network. It retains the rational-frame contract and the compact-control movement proof. The complex network and its guard constants are unchanged. The result remains conditional on the pinned upstream manuscript and the previously identified compact-control and Gaussian extensions.

Zhihao Chen's earlier PR #7 introduces the same ternary five-subset motif, with a different producer and a stronger claimed witness. The circuit below is a separate implementation; no priority claim for the motif or verification of that pending PR is made.

2.1 The two fields and the side map

Let $h = 29$, let $\mathcal{T} = \binom{[h]}{5}$, and put $n = \binom{h}{5} = 118755$, $r = \binom{h}{2} = 406$. Over $\mathbb{F}_3$ let B be the incidence matrix of five-subsets against pairs. Then

$$C = BB^\top, \quad C_{ST} = \binom{|S \cap T|}{2} \pmod{3}.$$

For $t = 0, 1, 2, 3, 4, 5$ the binomial residues are $0, 0, 1, 0, 0, 1$. Consequently $C = I + A_2$, where A_2 is the adjacency matrix for intersection exactly two. The side map is $-A_2$; thus the central and side maps add to the identity. Only the displayed factor size r is needed, not a claim that it is a minimal factor.

The rational label space is $F = \mathbb{Q}^h$ with form

$$H = I - \frac{2}{25}J.$$

It is nondegenerate, since $1 - 2h/25 = -33/25 \neq 0$. Write t_T for the indicator of T . Then

$$t_S^\top H t_T = |S \cap T| - 2, \quad t_T^\top H t_T = 3.$$

Let P_T be projection onto $\langle t_T \rangle$ in this form. Intersection-two neighbors have mutually annihilating projections. The scalar characteristic and rational label field are different; the address prime in the matrix-shear procedure is a third, independently chosen fixed prime avoiding its rational denominators.

2.2 An aligned side sum circuit

For each pair $J \subset [h]$, take all source variables $x_{J \cup S}$ with $|S| = 3$ and $S \cap J = \emptyset$. For each triple T disjoint from J compute

$$z_{J,T} = \sum_{S \cap (J \cup T) = \emptyset, |S|=3} x_{J \cup S}.$$

Inject $-z_{J,T}$ into target $J \cup T$. An ordered intersection-two pair has the unique common pair J , so these partial outputs implement $-A_2$ exactly. There are $q = \binom{h}{2} \binom{h-2}{3} = 10n$ output uses.

Use the deterministic disjoint-triple sum DAG of the tensor construction on all h points. In group J , delete formal source variables meeting J and retain only output triples disjoint from J . A zero in this substitution denotes an absent formal input, not a cleared physical scratch register. Every retained addition still has disjoint source supports. The fixed global partition order aligns intermediate sums between different common-pair groups.

For completeness, the tensor sum recipe is a finite recursion on ordered ground sets. Let $Z(n, a, b)$ be its unintermed addition count for summing a -subset inputs over all b -subset disjointness queries. If $a + b > n$, the outputs are empty sums; if $a = 0$, copy the sole input. Both have zero cost. If $b = 0$, sum the $\binom{n}{a}$ inputs at cost $\binom{n}{a} - 1$. Otherwise start with separate query sums, costing $\binom{n}{b} (\binom{n-b}{a} - 1)$. For each $1 \leq l \leq \lfloor n/2 \rfloor$, put $r_0 = n - l$ and split the ground set into its first l and last r_0 points. For every feasible input split i and query split j , contract one factor and then the other, at cost

$$\min \left\{ \binom{r_0}{a-i} Z(l, i, j) + \binom{l}{j} Z(r_0, a-i, b-j), \quad \binom{l}{i} Z(r_0, a-i, b-j) + \binom{r_0}{b-j} Z(l, i, j) \right\}.$$

Feasibility requires the subset sizes to fit their factors and $i + j \leq l$, $a - i + b - j \leq r_0$. For each j , add the resulting terms for each full query: if there are $t_j > 0$ feasible i , this costs $\binom{l}{j} \binom{r_0}{b-j} (t_j - 1)$. Choose a split only if its total strictly improves the current best, testing l in increasing order. For the displayed minimum prefer its first contraction order on a tie. Use lexicographic subset order and increasing split indices; sum lists by recursively splitting them into two halves. Intern every equal formal support at each addition, retaining its first decomposition. The initial template is the resulting $(n, a, b) = (h, 3, 3)$ DAG. This describes a cancellation-free integer coefficient computation: disjointness factors across the two ground sets, and the input split partitions the sources.

Within a group, intern equal full formal support sets. Across groups, let I be the intersection of all full five-subset source labels of a nontrivial sum. A sum shared by distinct groups has $|I| \geq 3$. If $|I| = 3$, its exact signature is I and the set of residual pairs $S \setminus I$; if $|I| = 4$, use I and its residual points. These signatures reconstruct the full set of five-subset variables, so there are no hash collisions at the mathematical level. A nontrivial sum cannot have $|I| = 5$. Sums with $|I| = 2$ belong only to that common-pair group. Equal singleton inputs are identified by their five-subset labels. Retain the first decomposition of each sum and prune unused nodes after all groups are processed.

The exact integer DAG-array construction at $h = 29$ gives

$$c = 19593239, \quad q = 1187550, \quad S = c + q = 20780789.$$

Here c counts active additions after pruning, not allocated nodes. The implementation independently expands every support and output row at $h = 8, 9$, checks the signature identifications there, and regenerates the full integer DAG counts at $h = 29$. The general exact-signature argument above justifies the large sharing operation; the small tests are implementation controls, not an extrapolated proof.

2.3 Restoration and rational side frames

A cancellation-free addition DAG with c additions and q output uses has an invertible gather-and-fanout embedding on $S = c + q$ roles: each binary addition identifies one input pivot with an output use, while all other uses receive their own role. No initial side value is assumed zero. Let its invertible scalar operation be L , its data copy be V , and its signed output injection be J (coefficient -1 at every partial output). With central gather $G = B^\top$ and scatter $R = B$, use

$$L, -J, L^{-1}, -R, V, G, R, L, J, L^{-1}, -G, -V.$$

On arbitrary side input z , the two injections contribute $-JLz + JL(z + Vx) = JLVx = -A_2x$. The inverse mixers and final negative copy restore z . Central input cancels in the same way, leaving $RGx = Cx$. Thus $y \leftarrow y + x$ and every auxiliary input is restored. The actual reverse inverse schedule gives $y \leftarrow y - x$.

Every nonzero vector u in the span of five-subset indicators containing one fixed pair $\{a, b\}$ satisfies

$$u_a = u_b, \quad \sum_i u_i = 5u_a, \quad u^\top H u = \sum_{i \notin \{a, b\}} u_i^2 > 0.$$

If the last sum vanished, the first two identities would give $2u_a = 5u_a$, hence $u = 0$. Therefore every source span appearing in an original group is nondegenerate. Interning an equal formal sum does not change its span, so this remains true after cross-group sharing.

For a DAG node v choose the orthogonal projection E_v onto its source span. These projections increase along forward edges. Every ancestor source is orthogonal to every reachable target, so

the final projection at a partial output lies below $I - P_T$ for its target T . For the reverse mixer use $I - E_v$. Orthogonal complementation reverses the inclusions and gives increasing reverse edges, including the target lines at entry and source complements at exit. Rational nondegeneracy suffices; no characteristic-two orthonormal-residual assumption is used.

2.4 Complete tensor circuit and rank ledger

Index the data banks by $\mathcal{T}^3$, with $N = n^3$ and $m = h^3$. Use the first-coordinate shears, inverse second-coordinate shears with logical banks exchanged, and third-coordinate shears. The scalar action is $(X, Y) \mapsto (-Y, X)$. Negate the physical X bank at its final identity frame to obtain the unsigned bank exchange. This fixed pointwise gate has zero added edge rank and leaves all restored auxiliaries unchanged.

For a stage write $A = B_0 \perp P$, where P is its preceding label tensor line, and suppress the future tensor line Q . Put

$$D_0 = B_0 \otimes I, \quad D_1 = A \otimes I, \quad D(E) = D_0 + P \otimes E.$$

The common frames of the forward twelve operations are

$$D_0, D_0, D_0, D_0, D(P_T), D_1, D_0, D(E_v), D(I - P_T), D_1, D_1, D_1.$$

For the reversed inverse invocation they are

$$D_0, D_0, D_0, D(P_T), D(I - E_v), D_1, D_0, D(I - P_T), D_1, D_1, D_1, D_1.$$

Copy and injection gates are grouped by their data role, including all partial outputs for that target. Central gates retain every data and central incidence, including zero factor coefficients. All frames are tensored with Q . These are the same tensor boundaries as the general three-stage compiler. All side changes increase; only the central return decreases, by h on each of the r central roles in each invocation. Thus the total decreasing dimension is

$$L_{\text{dec}} = 3n^2rh.$$

The signed dimension telescoping gives $Wm - 2N + 2L_{\text{dec}}$. Replace the physical X source projections P_a by $-P_a$; because $2 \neq 0$ in the rational frame field, this adds exactly N to the rank sum. Source frames are now $-P_a, 0$, sink frames $I, I - P_a$, and each restored auxiliary has source 0, sink I . The unsigned scalar exchange therefore satisfies every endpoint difference I_m .

The bipartite double cover of the intersection-two graph is regular of positive degree $10\binom{h-5}{3}$. Hall's condition follows by edge counting, and a fixed deterministic matching algorithm supplies an orthogonal label permutation π . Share stage-one invocation (A, B) with stage-three invocation $(B, \pi(A))$, keeping local auxiliary indices. The boundary projections are

$$E = I \otimes P_A \otimes P_B, \quad H_0 = (I - P_B \otimes P_{\pi(A)}) \otimes I.$$

They satisfy $EH_0 = H_0E = E$. The twelve-operation schedule already has initial frame D_0 on every auxiliary, so the join replaces two boundary segments by one and saves exactly m ranks per identified role. Consequently

$$W = 2n^3 + 2n^2(S + r), \quad \Delta = Wm - s = n^3 - 6n^2rh.$$

Numerically,

$$\begin{aligned} m &= 24389, & N &= 1674772079218875, \\ W &= 589493540769997500, \\ L_{\text{dec}} &= 498137336383050, \\ \Delta &= 678497406452775, \\ s &= 14377157287342062574725. \end{aligned}$$

Writing $\eta = \Delta/(Wm)$, the exact logarithm enclosure verifies

$$\eta > \frac{467}{10^{11}} \log m.$$

Since $1 - \eta < e^{-\eta}$, this proves $s/W < m^{1-a_b}$ with $a_b = 467/10^{11}$. The finite-alphabet transfer therefore supplies the bit interchange exponent $\tau = 1 - 467/10^{11}$.

2.5 Conditional assembly witness

Retain the separately integrated complex network with $a_c = 5/10^9$, its arity 17576, and its existing node charge. Its stopped-depth constants remain $\beta = 1/100$, $\zeta = 1/1000$ and $C_1 = 4961/1000$. The bit alphabet extension is exact and introduces no complex rounding operations. Its fixed symbol width and circuit size alter constants only, not the complex precision recurrence.

Take

$$\epsilon = \frac{1999}{10000}, \quad c = 1, \quad \delta = 10^{-6}, \quad \lambda = 1 - \frac{4669}{10^{12}}, \quad \lambda' = 1 - \frac{4668}{10^{12}}, \quad \kappa = 2^{-30}.$$

Since $\sigma < \tau$, the compact-control internal exponent is $\chi = \tau$. The reserved-axis preprocessing exponent is zero. Exact rational arithmetic checks every compact-control, Gaussian, leaf, prefix, dimension, guard and assembly constraint, with these two proved substitutions for the old packed overhead. All seven final margins exceed κ ; the minimum is

$$g_3 = \frac{2332833}{2500000000000000}, \quad g_3 - 2^{-30} = \frac{296652863}{16384000000000000000} > 0.$$

Subject to the retained upstream and written extension hypotheses, this therefore supports $T(n) = O(n(\log n)^{1-2^{-30}})$. It is an asymptotic exponent comparison, not a practical runtime claim or independent verification of the upstream multiplication theorem.

3 Reproduction

The construction and exact conditional assembly are checked by `scripts/audit_ternary_side.py`. The complete independent manuscript patch is `patches/ternary-30.patch`, generated by `scripts/make_ternary_patch.py`. Its finite alphabet, separate network arities, retained movement proofs, parameter substitutions and internal references are checked by `tests/test_ternary_patch.py`. Earlier certificates, notes and patches are preserved. Finite controls do not replace the general construction proofs or independently verify upstream.